Enterprise AI Agentic eQMS Architecture
Google ADK (Java) Technical Design Document

Prepared based on the uploaded eQMS workflow diagrams covering Deviation, CAPA, Change Control, Complaint, Audit, Document Management, Equipment, Nonconformance, Risk, Supplier and Training workflows.

Chapter 1: Executive Summary

This chapter describes the architecture and implementation guidance for Executive Summary. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Executive Summary. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Executive Summary. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 2: Existing Workflow Assessment

This chapter describes the architecture and implementation guidance for Existing Workflow Assessment. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring

Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Existing Workflow Assessment. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Existing Workflow Assessment. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 3: Target AI Vision

This chapter describes the architecture and implementation guidance for Target AI Vision. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Target AI Vision. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Target AI Vision. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 4: Google ADK Java Overview

This chapter describes the architecture and implementation guidance for Google ADK Java Overview. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Google ADK Java Overview. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Google ADK Java Overview. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 5: Reference Architecture

This chapter describes the architecture and implementation guidance for Reference Architecture. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Reference Architecture. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring

Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Reference Architecture. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 6: Agent Orchestrator

This chapter describes the architecture and implementation guidance for Agent Orchestrator. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Agent Orchestrator. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Agent Orchestrator. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 7: Supervisor Agent

This chapter describes the architecture and implementation guidance for Supervisor Agent. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Supervisor Agent. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Supervisor Agent. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 8: Deviation Agent

This chapter describes the architecture and implementation guidance for Deviation Agent. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Deviation Agent. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Deviation Agent. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 9: CAPA Agent

This chapter describes the architecture and implementation guidance for CAPA Agent. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for CAPA Agent. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for CAPA Agent. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 10: Complaint Agent

This chapter describes the architecture and implementation guidance for Complaint Agent. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Complaint Agent. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring.

Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Complaint Agent. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 11: Audit Agent

This chapter describes the architecture and implementation guidance for Audit Agent. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Audit Agent. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Audit Agent. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 12: Document Agent

This chapter describes the architecture and implementation guidance for Document Agent. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Document Agent. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Document Agent. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 13: Training Agent

This chapter describes the architecture and implementation guidance for Training Agent. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Training Agent. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Training Agent. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 14: Supplier Agent

This chapter describes the architecture and implementation guidance for Supplier Agent. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Supplier Agent. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Supplier Agent. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 15: Equipment Agent

This chapter describes the architecture and implementation guidance for Equipment Agent. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Equipment Agent. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Equipment Agent. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 16: Calibration Agent

This chapter describes the architecture and implementation guidance for Calibration Agent. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Calibration Agent. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Calibration Agent. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 17: Nonconformance Agent

This chapter describes the architecture and implementation guidance for Nonconformance Agent. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Nonconformance Agent. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Nonconformance Agent. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 18: Risk Agent

This chapter describes the architecture and implementation guidance for Risk Agent. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Risk Agent. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Risk Agent. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and

explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 19: Change Control Agent

This chapter describes the architecture and implementation guidance for Change Control Agent. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Change Control Agent. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Change Control Agent. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 20: Regulatory Intelligence

This chapter describes the architecture and implementation guidance for Regulatory Intelligence. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Regulatory Intelligence. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Regulatory Intelligence. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 21: Knowledge Graph

This chapter describes the architecture and implementation guidance for Knowledge Graph. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Knowledge Graph. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Knowledge Graph. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 22: RAG Architecture

This chapter describes the architecture and implementation guidance for RAG Architecture. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for RAG Architecture. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for RAG Architecture. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 23: Vector Database

This chapter describes the architecture and implementation guidance for Vector Database. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Vector Database. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Vector Database. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 24: Prompt Engineering

This chapter describes the architecture and implementation guidance for Prompt Engineering. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Prompt Engineering. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Prompt Engineering. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 25: Tool Registry

This chapter describes the architecture and implementation guidance for Tool Registry. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Tool Registry. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with

Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Tool Registry. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 26: MCP Integration

This chapter describes the architecture and implementation guidance for MCP Integration. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for MCP Integration. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for MCP Integration. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 27: REST APIs

This chapter describes the architecture and implementation guidance for REST APIs. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for REST APIs. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for REST APIs. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 28: Spring Boot

This chapter describes the architecture and implementation guidance for Spring Boot. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Spring Boot. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Spring Boot. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 29: Event Driven Design

This chapter describes the architecture and implementation guidance for Event Driven Design. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Event Driven Design. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Event Driven Design. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 30: Kafka

This chapter describes the architecture and implementation guidance for Kafka. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Kafka. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Kafka. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 31: Security

This chapter describes the architecture and implementation guidance for Security. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Security. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Security. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 32: 21 CFR Part 11

This chapter describes the architecture and implementation guidance for 21 CFR Part 11. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for 21 CFR Part 11. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for 21 CFR Part 11. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 33: Annex 11

This chapter describes the architecture and implementation guidance for Annex 11. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Annex 11. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Annex 11. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 34: GAMP5 Validation

This chapter describes the architecture and implementation guidance for GAMP5 Validation. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for GAMP5 Validation. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for GAMP5 Validation. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 35: Audit Trail

This chapter describes the architecture and implementation guidance for Audit Trail. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Audit Trail. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Audit Trail. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 36: Human Approval

This chapter describes the architecture and implementation guidance for Human Approval. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Human Approval. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Human Approval. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 37: Memory

This chapter describes the architecture and implementation guidance for Memory. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with

Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Memory. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Memory. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 38: Observability

This chapter describes the architecture and implementation guidance for Observability. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Observability. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Observability. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and

explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 39: Dashboards

This chapter describes the architecture and implementation guidance for Dashboards. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Dashboards. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Dashboards. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 40: Predictive Compliance

This chapter describes the architecture and implementation guidance for Predictive Compliance. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Predictive Compliance. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Predictive Compliance. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 41: Root Cause AI

This chapter describes the architecture and implementation guidance for Root Cause AI. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Root Cause AI. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Root Cause AI. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 42: LLM Selection

This chapter describes the architecture and implementation guidance for LLM Selection. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for LLM Selection. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for LLM Selection. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 43: Deployment

This chapter describes the architecture and implementation guidance for Deployment. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Deployment. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Deployment. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and

explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 44: Kubernetes

This chapter describes the architecture and implementation guidance for Kubernetes. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Kubernetes. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Kubernetes. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 45: Cost Model

This chapter describes the architecture and implementation guidance for Cost Model. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Cost Model. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Cost Model. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 46: Implementation Roadmap

This chapter describes the architecture and implementation guidance for Implementation Roadmap. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Implementation Roadmap. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Implementation Roadmap. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 47: Testing

This chapter describes the architecture and implementation guidance for Testing. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Testing. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Testing. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 48: Future Enhancements

This chapter describes the architecture and implementation guidance for Future Enhancements. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Future Enhancements. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Future Enhancements. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring

Boot services, agent tools, event topics, security, validation strategy, and phased rollout.

Chapter 49: Appendix

This chapter describes the architecture and implementation guidance for Appendix. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Appendix. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout. This chapter describes the architecture and implementation guidance for Appendix. The solution uses Google ADK Java agents coordinated by a Supervisor Agent. Agents invoke enterprise tools through MCP and REST connectors, retrieve validated knowledge using RAG, maintain complete audit trails, and always keep human approvals for GxP decisions. Each workflow integrates with Deviation, CAPA, Document, Training, Supplier, Equipment, Complaint, Nonconformance, Risk and Change Control modules. Key capabilities include automatic classification, investigation assistance, draft generation, cross-module triggering, predictive analytics, compliance monitoring, evidence collection, inspection readiness, KPI generation, and explainable AI with confidence scoring. Implementation guidance covers Java packages, Spring Boot services, agent tools, event topics, security, validation strategy, and phased rollout.